

Visualizing the Central Limit Theorem Through Repeated Sampling of Housing Prices

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Objective

To analyse a large, real-life dataset and bridge the gap between lectures and practical application by visualizing the Central Limit Theorem and analysing the changes sample size makes on the overall distribution.

Dataset Description

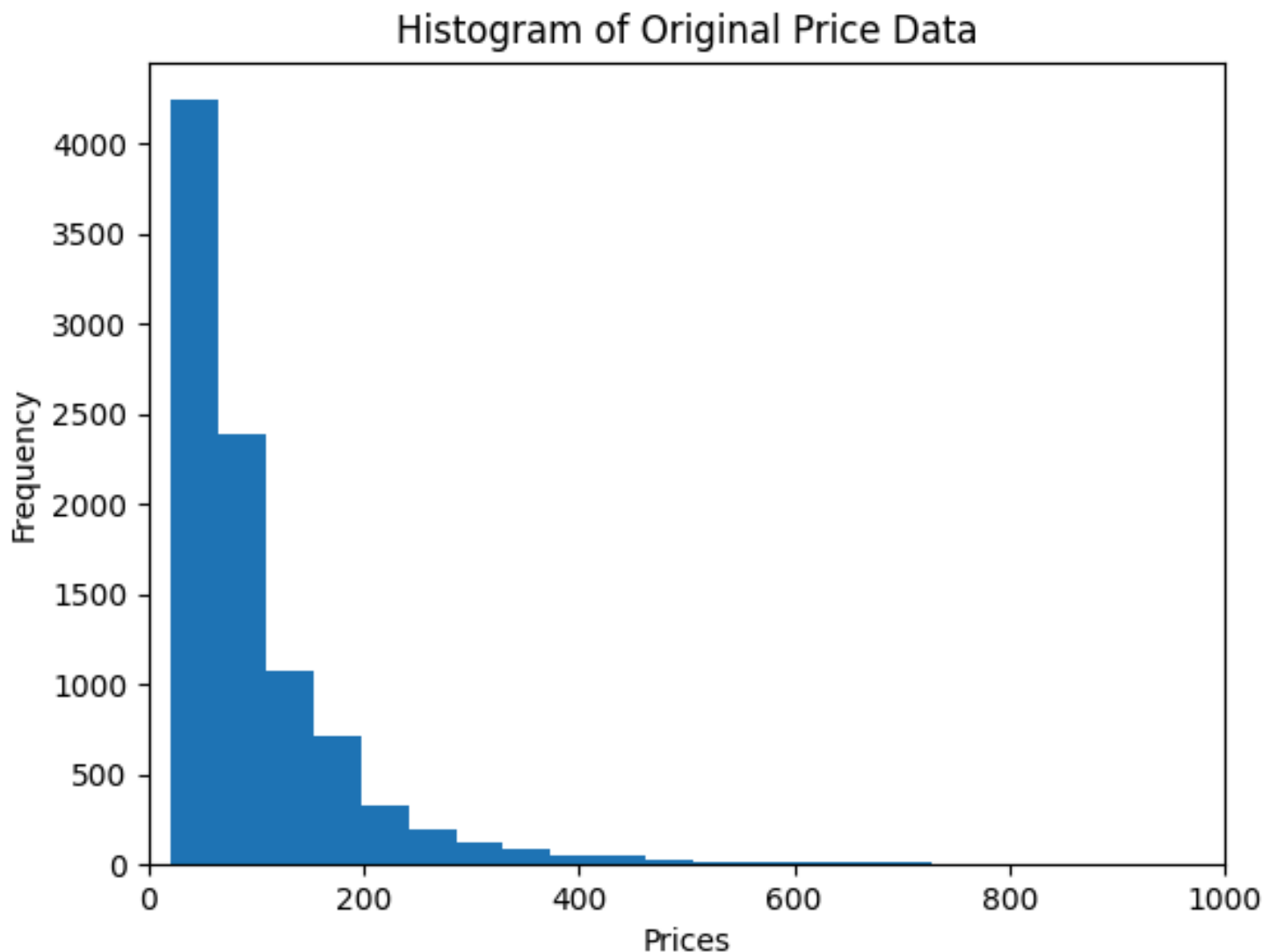
A .csv file containing information about **Indian House Prices** was used. Although only the **Prices** column was extracted and analysed.

Formulas Used

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s = \sqrt{s^2},$$
$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{N}}.$$

Original-Data Histogram



Part B: Discuss the Original Data Distribution

1. Is the original Price distribution symmetric or skewed?

Ans: Skewed.

2. Does the original Price distribution look normally distributed?

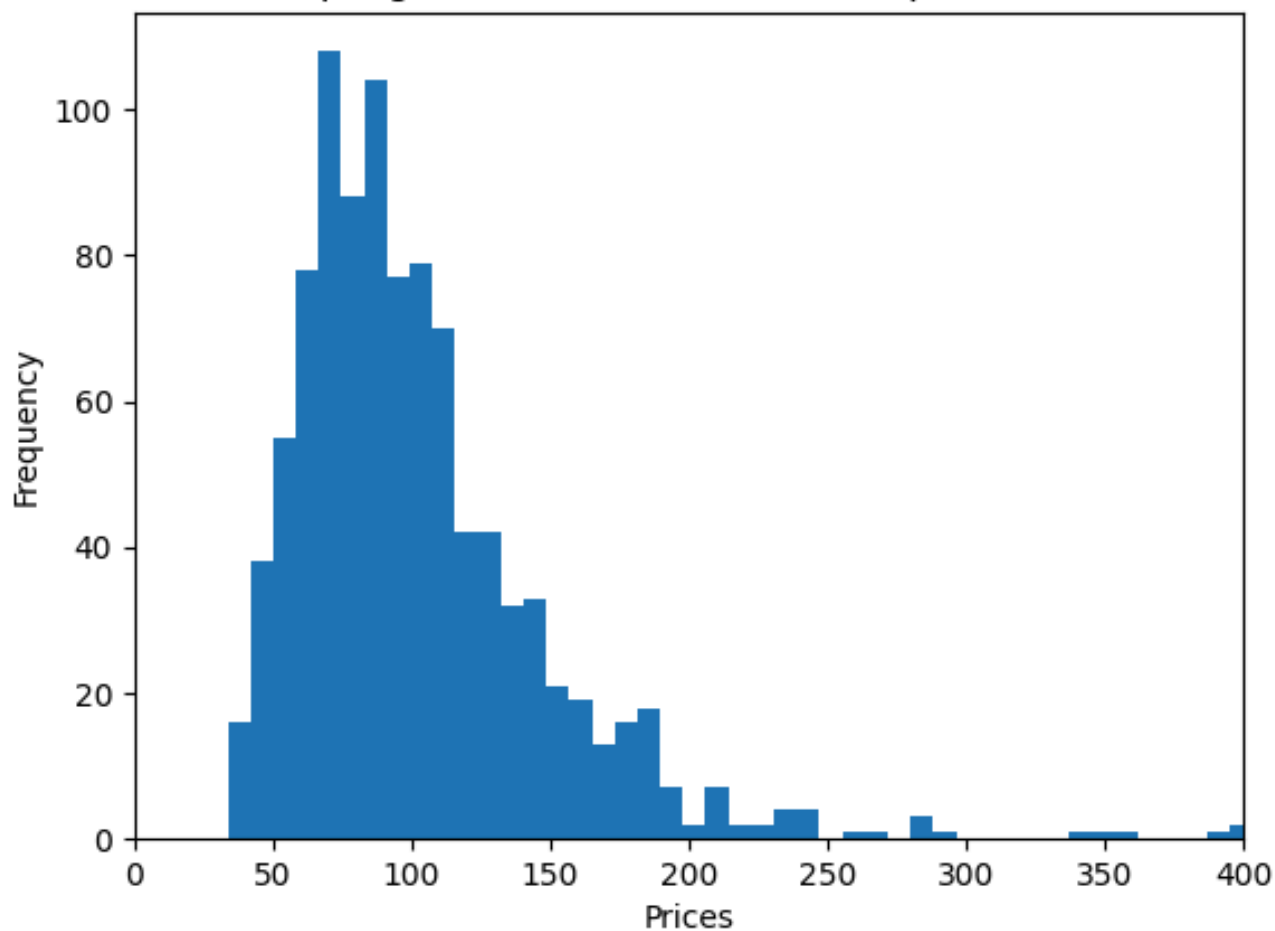
Ans: No, it does not look normally distributed.

3. Why might real-world housing-price data be skewed?

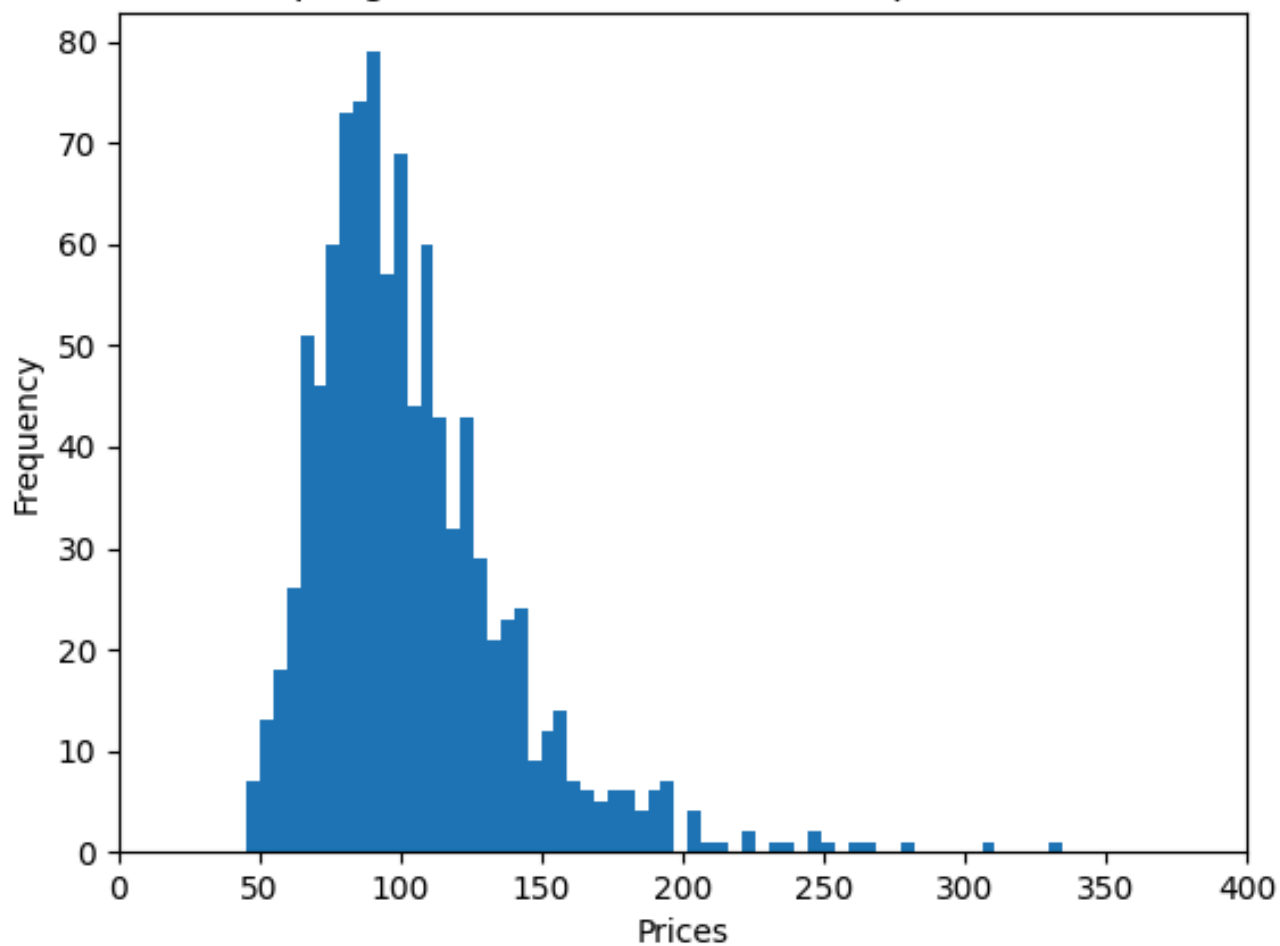
Ans: Because in the real-world only a minority of people can afford expensive houses and majority have cheaper houses hence the difference in frequency makes the distribution look skewed.

Histograms of Sample Means

Sampling Distribution of Mean (Sample size = 5)

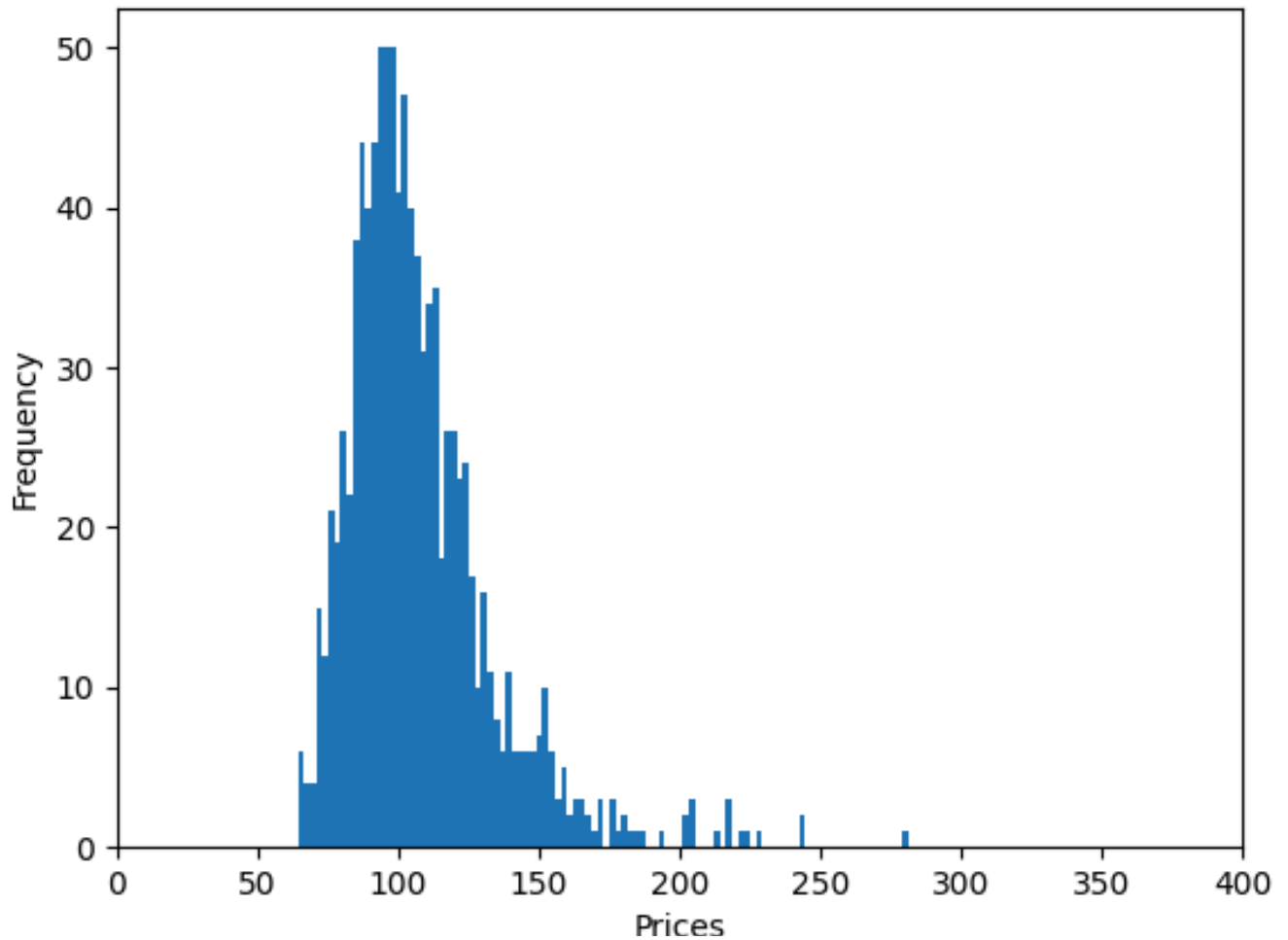


Sampling Distribution of Mean (Sample size = 10)

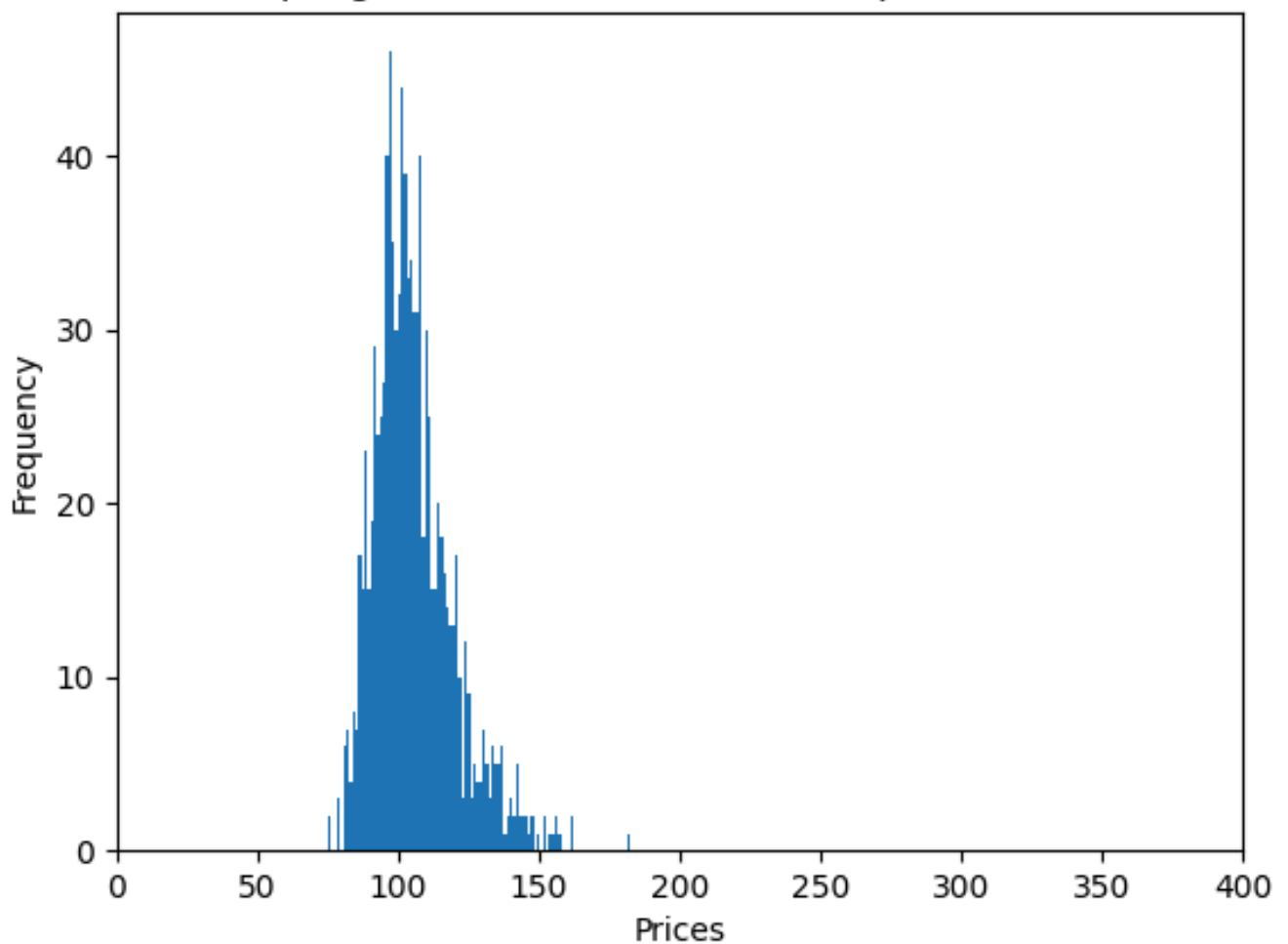


Histograms of Sample Means

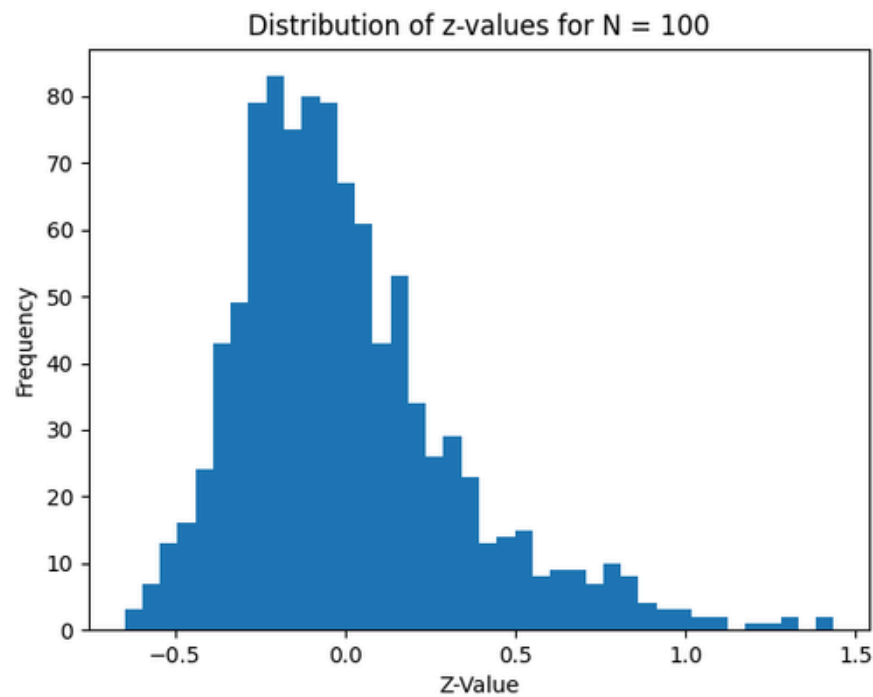
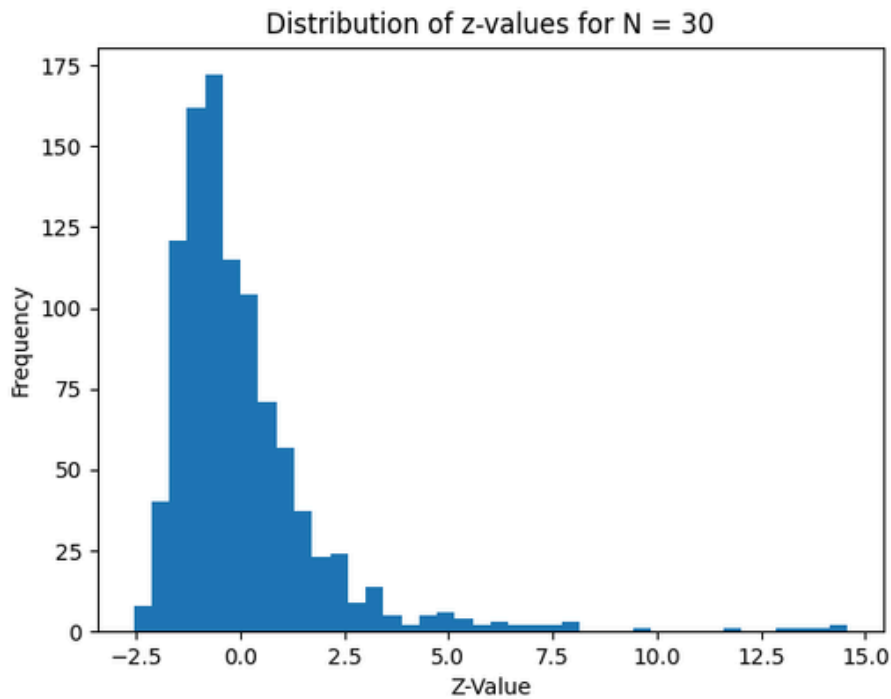
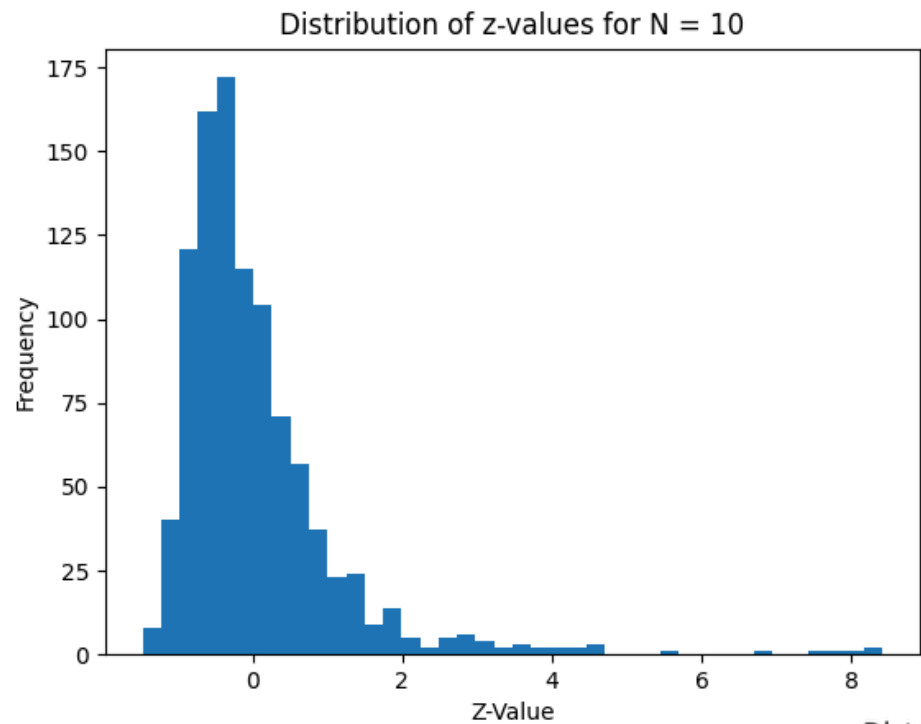
Sampling Distribution of Mean (Sample size = 30)



Sampling Distribution of Mean (Sample size = 100)



Histograms of Standardized Z-Values



Part E: Discussion and Interpretation

1. What happens to the spread of the sample means as N increases?

Ans: The spread of the mean gradually becomes closer and closer to a normal distribution.

2. What happens to the shape of the sampling distribution as N increases?

Ans: The shape becomes more symmetrical and bell-like.

3. Does the sampling distribution become more nearly normal even if the original Price data are not normal?

Ans: Yes, as the sample size increases the distribution becomes more normal proving CLT.

4. What is the role of the standard error $\sigma/\sqrt{N}$?

Ans: It essentially controls how close our answer is to the actual answer. A larger SE means SD is large, which means the spread is large, which means we are farther from the actual answer. Similarly, a larger sample size means a smaller error which means we are closer to the actual answer.

5. Why does increasing N make the sample mean more reliable?

Ans: Increasing N means taking more data, hence getting closer to the actual mean. A perfect answer would be achieved if $N = \text{infinite}$.

6. How does your simulation confirm the Central Limit Theorem?

Ans: CLT states that the distribution of means always follows normal distribution. We started out with a non-normal distribution, and through random samples we achieved nearly normal distribution. A data so random, as the 1st figure shows, can turn into normal if we take more and more samples proves the correctness of CLT.

Conclusion

To sum up, through the random sampling and plotting, we can conclude that the Central Limit Theorem does in fact states correctly that distribution of means follows a normal distribution.